SOURCE CODE :

```
import tkinter as tk
from tkinter import ttk, messagebox, simpledialog
import json
import os
from datetime import datetime


# ============================================================
# WATER DISTRIBUTION AND CONSUMPTION MANAGEMENT SYSTEM
# Mixed Data Structures:
# List + Tuple + Set + Dictionary
# ============================================================


# ------------------------------------------------------------
# GLOBAL DATA
# ------------------------------------------------------------

# LIST
# Each complete record is stored as a tuple inside this list.
records = []

# SET
# Used to maintain unique zones.
unique_zones = set()

# SET
# Used to maintain unique consumption classifications.
consumption_categories = set()

# DICTIONARY
# Stores a quick household/building-wise reference.
user_directory = {}

# DICTIONARY
# Stores zone-wise consumption totals.
zone_consumption = {}

DATA_FILE = "water_records.json"
REPORT_FILE = "water_analysis_report.txt"

# Consumption thresholds
LOW_LIMIT = 500
MODERATE_LIMIT = 1000
EXCESSIVE_LIMIT = 1000


# ------------------------------------------------------------
# RECORD FORMAT
```

```
# ------------------------------------------------------------
# Tuple structure:
#
# (
#   ID,
#   Zone,
#   Water Supplied,
#   Water Consumed,
#   Date,
#   Classification
# )
#
# Example:
# ("H001", "North", 1500.0, 700.0, "01-09-2026", "Moderate Consumption")
# ------------------------------------------------------------


# ------------------------------------------------------------
# VALIDATION FUNCTIONS
# ------------------------------------------------------------

def clean_text(value):
    """Remove unnecessary spaces from text."""
    return value.strip()


def validate_id(value):
    value = clean_text(value)

    if value == "":
        return False, "Household/Building ID cannot be empty."

    if len(value) < 2:
        return False, "ID must contain at least 2 characters."

    return True, value.upper()


def validate_zone(value):
    value = clean_text(value)

    if value == "":
        return False, "Zone cannot be empty."

    if not any(char.isalpha() for char in value):
        return False, "Zone must contain letters."

    return True, value.title()
```

```
def validate_quantity(value, field_name):
    try:
        number = float(value)

        if number < 0:
            return False, field_name + " cannot be negative."

        return True, number

    except ValueError:
        return False, field_name + " must be a valid number."


def validate_date(value):
    value = clean_text(value)

    try:
        datetime.strptime(value, "%d-%m-%Y")
        return True, value
    except ValueError:
        return False, "Date must be in DD-MM-YYYY format."


# ------------------------------------------------------------
# CONSUMPTION CLASSIFICATION
# ------------------------------------------------------------

def classify_consumption(amount):

    if amount <= LOW_LIMIT:
        return "Low Consumption"

    elif amount <= MODERATE_LIMIT:
        return "Moderate Consumption"

    else:
        return "High Consumption"


# ------------------------------------------------------------
# REBUILD DATA STRUCTURES
# ------------------------------------------------------------

def rebuild_indexes():

    global unique_zones
    global consumption_categories
    global user_directory
    global zone_consumption
```

```
    unique_zones = set()
    consumption_categories = set()
    user_directory = {}
    zone_consumption = {}

    for record in records:

        user_id, zone, supplied, consumed, date, category = record

        # SET
        unique_zones.add(zone)

        # SET
        consumption_categories.add(category)

        # DICTIONARY
        user_directory[user_id] = {
            "zone": zone,
            "supplied": supplied,
            "consumed": consumed,
            "date": date,
            "category": category
        }

        # DICTIONARY
        if zone not in zone_consumption:
            zone_consumption[zone] = 0

        zone_consumption[zone] += consumed


# ------------------------------------------------------------
# ADD RECORD
# ------------------------------------------------------------

def add_record():

    record_window = tk.Toplevel(root)
    record_window.title("Add Water Distribution Record")
    record_window.geometry("450x430")
    record_window.resizable(False, False)

    frame = ttk.Frame(record_window, padding=25)
    frame.pack(fill="both", expand=True)

    ttk.Label(
        frame,
        text="ADD WATER RECORD",
        font=("Arial", 18, "bold")
    ).pack(pady=(0, 20))
```

```
fields = {}

field_names = [
    ("Household / Building ID", "id"),
    ("Area / Zone", "zone"),
    ("Water Supplied (Litres)", "supplied"),
    ("Water Consumed (Litres)", "consumed"),
    ("Date (DD-MM-YYYY)", "date")
]

for label_text, key in field_names:

    row = ttk.Frame(frame)
    row.pack(fill="x", pady=7)

    ttk.Label(
        row,
        text=label_text,
        width=25
    ).pack(side="left")

    entry = ttk.Entry(row)
    entry.pack(side="right", expand=True, fill="x")

    fields[key] = entry

def save_new_record():

    valid, user_id = validate_id(fields["id"].get())

    if not valid:
        messagebox.showerror("Invalid ID", user_id)
        return

    valid, zone = validate_zone(fields["zone"].get())

    if not valid:
        messagebox.showerror("Invalid Zone", zone)
        return

    valid, supplied = validate_quantity(
        fields["supplied"].get(),
        "Water supplied"
    )

    if not valid:
        messagebox.showerror("Invalid Quantity", supplied)
        return
```

```
valid, consumed = validate_quantity(
    fields["consumed"].get(),
    "Water consumed"
)

if not valid:
    messagebox.showerror("Invalid Quantity", consumed)
    return

if consumed > supplied:
    messagebox.showerror(
        "Invalid Record",
        "Water consumed cannot be greater than water supplied."
    )
    return

valid, date = validate_date(fields["date"].get())

if not valid:
    messagebox.showerror("Invalid Date", date)
    return

# Check duplicate ID
for existing in records:

    if existing[0] == user_id:
        messagebox.showerror(
            "Duplicate ID",
            "This household/building ID already exists."
        )
        return

category = classify_consumption(consumed)

# TUPLE
new_record = (
    user_id,
    zone,
    supplied,
    consumed,
    date,
    category
)

# LIST
records.append(new_record)

rebuild_indexes()

save_data()
```

```
        update_dashboard()

        messagebox.showinfo(
            "Success",
            "Water record added successfully."
        )

        record_window.destroy()

    ttk.Button(
        frame,
        text="Save Record",
        command=save_new_record
    ).pack(pady=20)


# ------------------------------------------------------------
# DISPLAY RECORDS
# ------------------------------------------------------------

def display_records():

    window = tk.Toplevel(root)
    window.title("Water Distribution Records")
    window.geometry("950x520")

    ttk.Label(
        window,
        text="ALL WATER RECORDS",
        font=("Arial", 18, "bold")
    ).pack(pady=15)

    columns = (
        "ID",
        "Zone",
        "Supplied",
        "Consumed",
        "Date",
        "Classification"
    )

    tree = ttk.Treeview(
        window,
        columns=columns,
        show="headings"
    )

    for column in columns:
```

```
        tree.heading(column, text=column)

        tree.column(
            column,
            width=140,
            anchor="center"
        )

    tree.pack(
        fill="both",
        expand=True,
        padx=15,
        pady=10
    )

    for record in records:

        tree.insert(
            "",
            "end",
            values=(
                record[0],
                record[1],
                f"{record[2]:.2f} L",
                f"{record[3]:.2f} L",
                record[4],
                record[5]
            )
        )


# ------------------------------------------------------------
# SEARCH RECORDS
# ------------------------------------------------------------

def search_records():

    window = tk.Toplevel(root)
    window.title("Search Water Records")
    window.geometry("850x500")

    ttk.Label(
        window,
        text="SEARCH RECORDS",
        font=("Arial", 18, "bold")
    ).pack(pady=15)

    control = ttk.Frame(window)
    control.pack(pady=10)
```

```
ttk.Label(
    control,
    text="Search by:"
).pack(side="left", padx=5)

search_type = ttk.Combobox(
    control,
    values=["ID", "Zone", "Date"],
    state="readonly",
    width=12
)

search_type.current(0)
search_type.pack(side="left", padx=5)

search_entry = ttk.Entry(control, width=30)
search_entry.pack(side="left", padx=5)

columns = (
    "ID",
    "Zone",
    "Supplied",
    "Consumed",
    "Date",
    "Classification"
)

tree = ttk.Treeview(
    window,
    columns=columns,
    show="headings"
)

for column in columns:

    tree.heading(column, text=column)
    tree.column(column, width=125, anchor="center")

tree.pack(
    fill="both",
    expand=True,
    padx=15,
    pady=10
)

def perform_search():

    for item in tree.get_children():
        tree.delete(item)
```

```
        keyword = clean_text(search_entry.get()).lower()

        if keyword == "":
            messagebox.showwarning(
                "Search",
                "Enter something to search."
            )
            return

        option = search_type.get()

        for record in records:

            if option == "ID":
                value = record[0]

            elif option == "Zone":
                value = record[1]

            else:
                value = record[4]

            if keyword in value.lower():

                tree.insert(
                    "",
                    "end",
                    values=(
                        record[0],
                        record[1],
                        f"{record[2]:.2f} L",
                        f"{record[3]:.2f} L",
                        record[4],
                        record[5]
                    )
                )

    ttk.Button(
        control,
        text="Search",
        command=perform_search
    ).pack(side="left", padx=5)


# ------------------------------------------------------------
# UPDATE RECORD
# ------------------------------------------------------------

def update_record():
```

```
if not records:

    messagebox.showinfo(
        "Update",
        "No records available."
    )
    return

user_id = simpledialog.askstring(
    "Update Record",
    "Enter Household/Building ID:"
)

if not user_id:
    return

user_id = user_id.strip().upper()

index = -1

for i, record in enumerate(records):

    if record[0] == user_id:
        index = i
        break

if index == -1:

    messagebox.showerror(
        "Not Found",
        "Record not found."
    )
    return

old_record = records[index]

new_zone = simpledialog.askstring(
    "Update",
    "Enter zone:",
    initialvalue=old_record[1]
)

new_supplied = simpledialog.askstring(
    "Update",
    "Enter water supplied:",
    initialvalue=str(old_record[2])
)

new_consumed = simpledialog.askstring(
    "Update",
```

```
    "Enter water consumed:",
    initialvalue=str(old_record[3])
)

new_date = simpledialog.askstring(
    "Update",
    "Enter date DD-MM-YYYY:",
    initialvalue=old_record[4]
)

valid, zone = validate_zone(new_zone)

if not valid:
    messagebox.showerror("Error", zone)
    return

valid, supplied = validate_quantity(
    new_supplied,
    "Water supplied"
)

if not valid:
    messagebox.showerror("Error", supplied)
    return

valid, consumed = validate_quantity(
    new_consumed,
    "Water consumed"
)

if not valid:
    messagebox.showerror("Error", consumed)
    return

if consumed > supplied:

    messagebox.showerror(
        "Error",
        "Consumption cannot exceed supplied water."
    )
    return

valid, date = validate_date(new_date)

if not valid:
    messagebox.showerror("Error", date)
    return

category = classify_consumption(consumed)
```

```
    # Replace tuple in LIST
    records[index] = (
        user_id,
        zone,
        supplied,
        consumed,
        date,
        category
    )

    rebuild_indexes()
    save_data()
    update_dashboard()

    messagebox.showinfo(
        "Updated",
        "Record updated successfully."
    )


# ------------------------------------------------------------
# CLASSIFICATION VIEW
# ------------------------------------------------------------

def show_classification():

    window = tk.Toplevel(root)
    window.title("Consumption Classification")
    window.geometry("800x500")

    ttk.Label(
        window,
        text="CONSUMPTION CLASSIFICATION",
        font=("Arial", 18, "bold")
    ).pack(pady=15)

    columns = (
        "ID",
        "Zone",
        "Consumption",
        "Category"
    )

    tree = ttk.Treeview(
        window,
        columns=columns,
        show="headings"
    )

    for column in columns:
```

```
        tree.heading(column, text=column)
        tree.column(column, width=180, anchor="center")

    tree.pack(
        fill="both",
        expand=True,
        padx=20,
        pady=10
    )

    for record in records:

        tree.insert(
            "",
            "end",
            values=(
                record[0],
                record[1],
                f"{record[3]:.2f} L",
                record[5]
            )
        )


# ------------------------------------------------------------
# ANALYSIS CALCULATIONS
# ------------------------------------------------------------

def calculate_analysis():

    total_supplied = sum(
        record[2] for record in records
    )

    total_consumed = sum(
        record[3] for record in records
    )

    remaining = total_supplied - total_consumed

    if records:

        average = total_consumed / len(records)

    else:

        average = 0

    if total_supplied > 0:
```

```
        percentage = (
            total_consumed / total_supplied
        ) * 100

    else:

        percentage = 0

    high_users = [
        record for record in records
        if record[3] > EXCESSIVE_LIMIT
    ]

    return (
        total_supplied,
        total_consumed,
        remaining,
        average,
        percentage,
        high_users
    )


# ------------------------------------------------------------
# WATER ANALYSIS
# ------------------------------------------------------------

def show_analysis():

    window = tk.Toplevel(root)
    window.title("Water Analysis")
    window.geometry("700x600")

    ttk.Label(
        window,
        text="WATER ANALYSIS",
        font=("Arial", 20, "bold")
    ).pack(pady=20)

    (
        total_supplied,
        total_consumed,
        remaining,
        average,
        percentage,
        high_users
    ) = calculate_analysis()

    values = [
```

```
    ("Total Water Supplied", total_supplied),
    ("Total Water Consumed", total_consumed),
    ("Remaining Water", remaining),
    ("Average Consumption", average),
    ("Consumption Percentage", percentage)
]

for name, value in values:

    card = ttk.Frame(
        window,
        relief="ridge",
        borderwidth=2
    )

    card.pack(
        fill="x",
        padx=50,
        pady=8
    )

    if name == "Consumption Percentage":

        display_value = f"{value:.2f}%"

    else:

        display_value = f"{value:.2f} L"

    ttk.Label(
        card,
        text=name,
        font=("Arial", 11)
    ).pack(
        side="left",
        padx=15,
        pady=15
    )

    ttk.Label(
        card,
        text=display_value,
        font=("Arial", 12, "bold")
    ).pack(
        side="right",
        padx=15
    )

ttk.Label(
    window,
```

```
        text=f"High Consumption Users: {len(high_users)}",
        font=("Arial", 13, "bold")
    ).pack(pady=20)


# ------------------------------------------------------------
# EXCESSIVE CONSUMPTION
# ------------------------------------------------------------

def excessive_consumption():

    window = tk.Toplevel(root)
    window.title("Excessive Consumption")
    window.geometry("850x500")

    ttk.Label(
        window,
        text="EXCESSIVE WATER CONSUMPTION",
        font=("Arial", 18, "bold")
    ).pack(pady=15)

    ttk.Label(
        window,
        text=f"Consumption limit: {EXCESSIVE_LIMIT} L"
    ).pack()

    columns = (
        "ID",
        "Zone",
        "Consumed",
        "Date",
        "Category"
    )

    tree = ttk.Treeview(
        window,
        columns=columns,
        show="headings"
    )

    for column in columns:

        tree.heading(column, text=column)
        tree.column(column, width=150, anchor="center")

    tree.pack(
        fill="both",
        expand=True,
        padx=20,
        pady=20
```

```
    )

    found = False

    for record in records:

        if record[3] > EXCESSIVE_LIMIT:

            found = True

            tree.insert(
                "",
                "end",
                values=(
                    record[0],
                    record[1],
                    f"{record[3]:.2f} L",
                    record[4],
                    record[5]
                )
            )

    if not found:

        ttk.Label(
            window,
            text="No excessive consumption detected."
        ).pack(pady=10)


# ------------------------------------------------------------
# ZONE-WISE ANALYSIS
# ------------------------------------------------------------

def zone_analysis():

    rebuild_indexes()

    window = tk.Toplevel(root)
    window.title("Zone-wise Analysis")
    window.geometry("700x500")

    ttk.Label(
        window,
        text="ZONE-WISE WATER CONSUMPTION",
        font=("Arial", 18, "bold")
    ).pack(pady=20)

    columns = (
        "Zone",
```

```
        "Total Consumption",
        "Users"
    )

    tree = ttk.Treeview(
        window,
        columns=columns,
        show="headings"
    )

    for column in columns:

        tree.heading(column, text=column)
        tree.column(column, width=200, anchor="center")

    tree.pack(
        fill="both",
        expand=True,
        padx=30,
        pady=20
    )

    for zone in sorted(zone_consumption):

        users = 0

        for record in records:

            if record[1] == zone:
                users += 1

        tree.insert(
            "",
            "end",
            values=(
                zone,
                f"{zone_consumption[zone]:.2f} L",
                users
            )
        )


# ------------------------------------------------------------
# UNIQUE DATA ANALYSIS
# ------------------------------------------------------------

def unique_data():

    rebuild_indexes()
```

```
    window = tk.Toplevel(root)
    window.title("Unique Data Analysis")
    window.geometry("650x500")

    ttk.Label(
        window,
        text="UNIQUE DATA ANALYSIS",
        font=("Arial", 18, "bold")
    ).pack(pady=20)

    # SETS
    user_ids = {
        record[0] for record in records
    }

    zones = {
        record[1] for record in records
    }

    categories = {
        record[5] for record in records
    }

    information = [
        ("Unique Users / Buildings", user_ids),
        ("Unique Zones", zones),
        ("Consumption Categories", categories)
    ]

    for title, data in information:

        ttk.Label(
            window,
            text=title,
            font=("Arial", 12, "bold")
        ).pack(anchor="w", padx=30, pady=(15, 5))

        text = ", ".join(sorted(data)) if data else "None"

        ttk.Label(
            window,
            text=text,
            wraplength=550
        ).pack(anchor="w", padx=45)


# ---------------------------------------------------------
# SAVE DATA
# ---------------------------------------------------------
```

```
def save_data():

    try:

        # Convert tuples to lists because JSON does not
        # have a native tuple type.
        serializable_records = [
            list(record) for record in records
        ]

        with open(
            DATA_FILE,
            "w",
            encoding="utf-8"
        ) as file:

            json.dump(
                serializable_records,
                file,
                indent=4
            )

        return True

    except Exception as error:

        messagebox.showerror(
            "Save Error",
            str(error)
        )

        return False


# ------------------------------------------------------------
# LOAD DATA
# ------------------------------------------------------------

def load_data():

    global records

    if not os.path.exists(DATA_FILE):
        return

    try:

        with open(
            DATA_FILE,
            "r",
```

```
            encoding="utf-8"
        ) as file:

            stored_records = json.load(file)

        records = []

        # Convert loaded lists back into TUPLES.
        for item in stored_records:

            records.append(
                (
                    item[0],
                    item[1],
                    float(item[2]),
                    float(item[3]),
                    item[4],
                    item[5]
                )
            )

        rebuild_indexes()

    except Exception as error:

        messagebox.showerror(
            "Load Error",
            str(error)
        )


# ------------------------------------------------------------
# GENERATE ANALYTICAL REPORT
# ------------------------------------------------------------

def generate_report():

    if not records:

        messagebox.showinfo(
            "Report",
            "No records available to generate a report."
        )
        return

    (
        total_supplied,
        total_consumed,
        remaining,
        average,
```

```
    percentage,
    high_users
) = calculate_analysis()

rebuild_indexes()

report = []

report.append(
    "WATER DISTRIBUTION AND CONSUMPTION ANALYTICAL REPORT"
)

report.append("=" * 60)

report.append(
    f"Generated on: {datetime.now().strftime('%d-%m-%Y %H:%M:%S')}"
)

report.append("")

report.append("OVERALL ANALYSIS")
report.append("-" * 30)

report.append(
    f"Total Water Supplied   : {total_supplied:.2f} L"
)

report.append(
    f"Total Water Consumed   : {total_consumed:.2f} L"
)

report.append(
    f"Remaining Water        : {remaining:.2f} L"
)

report.append(
    f"Average Consumption    : {average:.2f} L"
)

report.append(
    f"Consumption Percentage : {percentage:.2f}%"
)

report.append("")

report.append("ZONE-WISE ANALYSIS")
report.append("-" * 30)

for zone in sorted(zone_consumption):
```

```
    report.append(
        f"{zone}: {zone_consumption[zone]:.2f} L"
    )

report.append("")

report.append("EXCESSIVE CONSUMPTION")
report.append("-" * 30)

if high_users:

    for record in high_users:

        report.append(
            f"{record[0]} | "
            f"{record[1]} | "
            f"{record[3]:.2f} L"
        )

else:

    report.append("No excessive consumption detected.")

report.append("")

report.append("UNIQUE ZONES")
report.append("-" * 30)

report.append(
    ", ".join(sorted(unique_zones))
)

report.append("")

report.append("UNIQUE CONSUMPTION CATEGORIES")
report.append("-" * 30)

report.append(
    ", ".join(sorted(consumption_categories))
)

report.append("")
report.append("=" * 60)

try:

    with open(
        REPORT_FILE,
        "w",
        encoding="utf-8"
```

```
        ) as file:

            file.write("\n".join(report))

        messagebox.showinfo(
            "Report Generated",
            f"Analytical report saved as:\n{REPORT_FILE}"
        )

    except Exception as error:

        messagebox.showerror(
            "Report Error",
            str(error)
        )


# ------------------------------------------------------------
# DASHBOARD UPDATE
# ------------------------------------------------------------

def update_dashboard():

    (
        total_supplied,
        total_consumed,
        remaining,
        average,
        percentage,
        high_users
    ) = calculate_analysis()

    supplied_value.config(
        text=f"{total_supplied:,.0f} L"
    )

    consumed_value.config(
        text=f"{total_consumed:,.0f} L"
    )

    remaining_value.config(
        text=f"{remaining:,.0f} L"
    )

    percentage_value.config(
        text=f"{percentage:.1f}%"
    )

    record_count_value.config(
        text=str(len(records))
```

```
    )

    high_count_value.config(
        text=str(len(high_users))
    )


# ------------------------------------------------------------
# EXIT
# ------------------------------------------------------------

def exit_application():

    save_data()
    root.destroy()


# ============================================================
# MAIN GUI
# ============================================================

root = tk.Tk()

root.title(
    "AquaTrack - Water Distribution & Consumption Management"
)

root.geometry("1100x720")

root.minsize(950, 650)


# ------------------------------------------------------------
# STYLE
# ------------------------------------------------------------

style = ttk.Style()

try:
    style.theme_use("clam")
except:
    pass

style.configure(
    "Title.TLabel",
    font=("Arial", 24, "bold")
)

style.configure(
    "Subtitle.TLabel",
```

```
    font=("Arial", 11)
)

style.configure(
    "Dashboard.TButton",
    font=("Arial", 11, "bold"),
    padding=12
)


# ------------------------------------------------------------
# HEADER
# ------------------------------------------------------------

header = ttk.Frame(
    root,
    padding=20
)

header.pack(fill="x")

ttk.Label(
    header,
    text="AquaTrack",
    style="Title.TLabel"
).pack()

ttk.Label(
    header,
    text="Water Distribution & Consumption Management System",
    style="Subtitle.TLabel"
).pack(pady=5)


# ------------------------------------------------------------
# SUMMARY CARDS
# ------------------------------------------------------------

summary = ttk.Frame(root)

summary.pack(
    fill="x",
    padx=25,
    pady=15
)


def create_card(parent, title):

    card = ttk.Frame(
```

```
        parent,
        relief="ridge",
        borderwidth=2,
        padding=15
    )

    card.pack(
        side="left",
        expand=True,
        fill="both",
        padx=5
    )

    ttk.Label(
        card,
        text=title,
        font=("Arial", 10)
    ).pack()

    value = ttk.Label(
        card,
        text="0",
        font=("Arial", 18, "bold")
    )

    value.pack(pady=8)

    return value


supplied_value = create_card(
    summary,
    "TOTAL SUPPLIED"
)

consumed_value = create_card(
    summary,
    "TOTAL CONSUMED"
)

remaining_value = create_card(
    summary,
    "REMAINING"
)

percentage_value = create_card(
    summary,
    "CONSUMPTION"
)
```

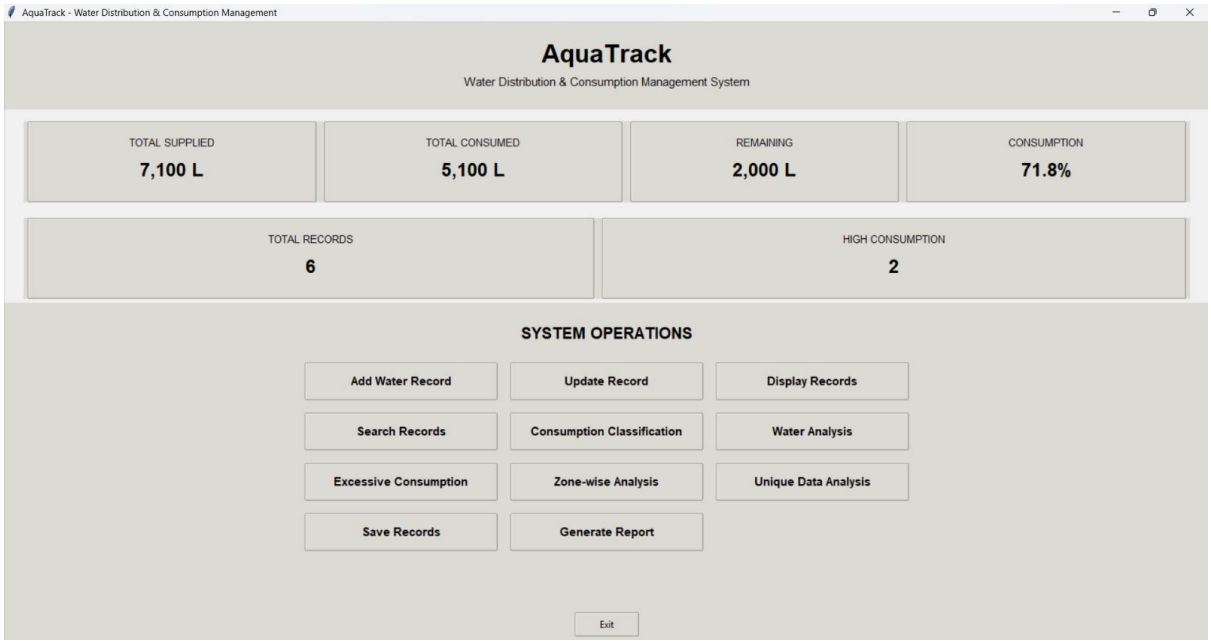
```
# ------------------------------------------------------------
# SECONDARY SUMMARY
# ------------------------------------------------------------

secondary = ttk.Frame(root)

secondary.pack(
    fill="x",
    padx=25,
    pady=5
)

record_count_value = create_card(
    secondary,
    "TOTAL RECORDS"
)

high_count_value = create_card(
    secondary,
    "HIGH CONSUMPTION"
)


# ------------------------------------------------------------
# MENU BUTTONS
# ------------------------------------------------------------

menu_frame = ttk.Frame(
    root,
    padding=20
)

menu_frame.pack(
    fill="both",
    expand=True
)

ttk.Label(
    menu_frame,
    text="SYSTEM OPERATIONS",
    font=("Arial", 15, "bold")
).pack(pady=(5, 15))


button_grid = ttk.Frame(menu_frame)

button_grid.pack()
```

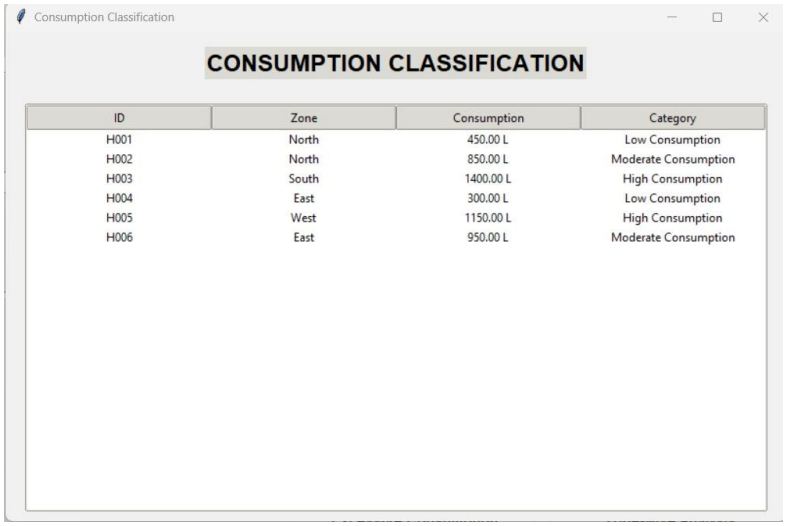
```
buttons = [
    ("Add Water Record", add_record),
    ("Update Record", update_record),
    ("Display Records", display_records),
    ("Search Records", search_records),
    ("Consumption Classification", show_classification),
    ("Water Analysis", show_analysis),
    ("Excessive Consumption", excessive_consumption),
    ("Zone-wise Analysis", zone_analysis),
    ("Unique Data Analysis", unique_data),
    ("Save Records", save_data),
    ("Generate Report", generate_report)
]


for index, (text, command) in enumerate(buttons):

    row = index // 3
    column = index % 3

    ttk.Button(
        button_grid,
        text=text,
        command=command,
        style="Dashboard.TButton",
        width=27
    ).grid(
        row=row,
        column=column,
        padx=8,
        pady=8
    )


# ------------------------------------------------------------
# FOOTER
# ------------------------------------------------------------

footer = ttk.Frame(
    root,
    padding=10
)

footer.pack(fill="x")

ttk.Button(
    footer,
    text="Exit",
    command=exit_application
).pack()
```

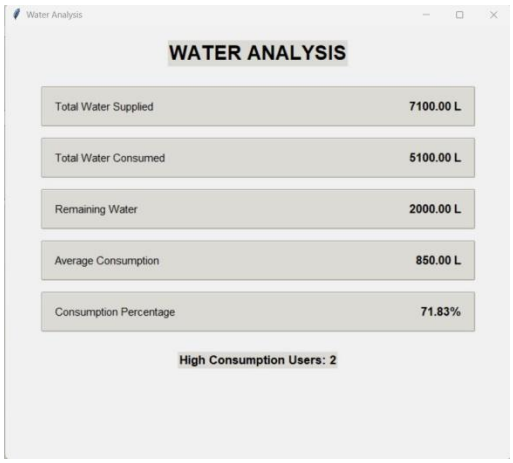
```
# ------------------------------------------------------
# LOAD PREVIOUS DATA
# ------------------------------------------------------

load_data()

update_dashboard()


# ------------------------------------------------------
# START APPLICATION
# ------------------------------------------------------

root.protocol(
    "WM_DELETE_WINDOW",
    exit_application
)

root.mainloop()
```

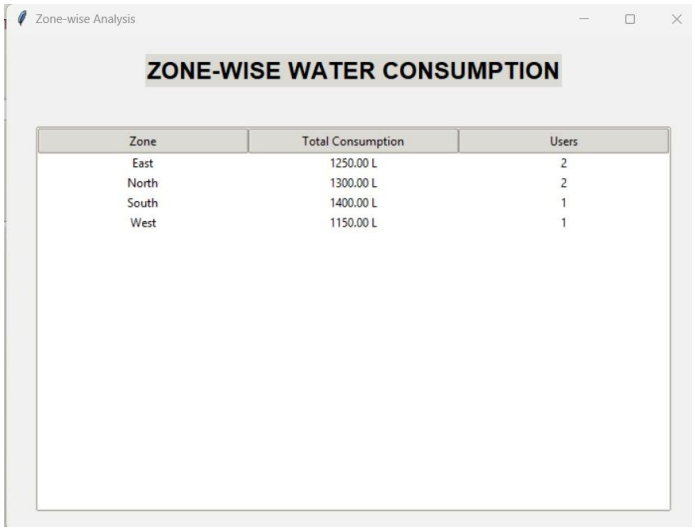
OUTPUT:

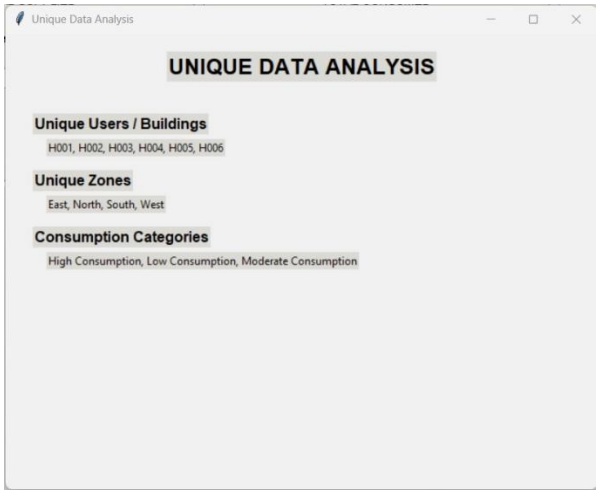
Display records:




Consumption Classification
CONSUMPTION CLASSIFICATION
ID
Zone
Consumption
Category
H001
North
450.00 L
Low Consumption
H002
North
850.00 L
Moderate Consumption
H003
South
1400.00 L
High Consumption
H004
East
300.00 L
Low Consumption
H005
West
1150.00 L
High Consumption
H006
East
950.00 L
Moderate Consumption


Water Analysis
WATER ANALYSIS
Total Water Supplied
7100.00 L
Total Water Consumed
5100.00 L
Remaining Water
2000.00 L
Average Consumption
850.00 L
Consumption Percentage
71.83%
High Consumption Users: 2


Zone-wise Analysis
ZONE-WISE WATER CONSUMPTION
Zone
Total Consumption
Users
East
1250.00 L
2
North
1300.00 L
2
South
1400.00 L
1
West
1150.00 L
1



Water analysis report:

File Edit View

```
WATER DISTRIBUTION AND CONSUMPTION ANALYTICAL REPORT
============================================================
Generated on: 02-09-2026 09:45:34

OVERALL ANALYSIS
------------------------------
Total Water Supplied   : 7100.00 L
Total Water Consumed   : 5100.00 L
Remaining Water        : 2000.00 L
Average Consumption    : 850.00 L
Consumption Percentage : 71.83%

ZONE-WISE ANALYSIS
------------------------------
East: 1250.00 L
North: 1300.00 L
South: 1400.00 L
West: 1150.00 L

EXCESSIVE CONSUMPTION
------------------------------
H003 | South | 1400.00 L
H005 | West | 1150.00 L

UNIQUE ZONES
------------------------------
East, North, South, West

UNIQUE CONSUMPTION CATEGORIES
------------------------------
High Consumption, Low Consumption, Moderate Consumption

============================================================
```